

BAHRAIN INTL, Bahrain

WMO: 411500

Lat: 26.2618N

Lon: 50.6433E

Elev: 2

StdP: 101.30

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.5	13.8	1.9	4.3	20.0	3.9	5.0	20.8	12.6	15.7	11.8	16.1	6.1	310	0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.9	41.2	23.7	40.1	24.5	39.1	25.1	31.2	35.5	30.7	35.2	30.1	34.9	4.5	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.1	27.4	34.6	29.5	26.4	34.4	29.0	25.6	34.2	106.4	35.7	103.7	34.9	100.8	34.7	34.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	10.2	9.3	DB	10.3	44.6	1.7	1.0	9.1	45.3	8.1	45.9	7.2	46.5	6.0	47.2	(4)
(5)			WB	6.7	31.9	2.0	0.9	5.3	32.5	4.1	33.1	3.0	33.6	1.6	34.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	17.8	18.8	22.0	26.6	31.8	34.4	35.7	35.7	33.7	30.3	24.8	19.8	(6)	
	DBStd	6.88	2.33	2.37	2.59	2.62	2.21	1.74	1.42	1.45	1.68	1.85	2.69	2.48	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	73	38	19	2	0	0	0	0	0	0	0	0	13	(9)	
	CDD10.0	6446	242	247	371	497	677	733	796	796	710	629	444	304	(10)	
	CDD18.3	3477	22	33	114	247	418	483	537	537	460	371	194	59	(11)	
	CDH23.3	48166	17	47	427	2239	6038	7806	8937	8876	7233	5008	1441	96	(12)	
(13)	CDH26.7	30517	1	5	100	773	3605	5406	6457	6396	4836	2597	337	4	(13)	
(14)	Wind	WSAvg	4.5	4.9	5.2	4.9	4.5	5.1	4.6	3.8	3.9	3.7	4.5	4.7	(14)	
(15) Precipitation	PrecAvg	58	16	15	9	7	1	0	0	0	0	1	6	10	(15)	
	PrecMax	166	88	107	72	52	10	0	5	1	0	30	62	96	(16)	
	PrecMin	9	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	40	19	23	14	11	2	0	1	0	0	4	13	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.1	27.0	32.8	36.8	41.1	42.8	43.2	42.8	40.8	37.2	32.8	27.1	(19)	
		MCWB	16.8	17.0	18.1	20.0	21.0	21.5	23.0	24.4	23.7	23.5	20.9	19.7	(20)	
	2%	DB	23.2	24.8	29.2	34.0	38.9	40.8	41.1	41.0	38.9	35.5	30.8	25.2	(21)	
		MCWB	17.2	16.9	18.0	20.2	21.4	22.2	24.0	25.6	25.3	24.2	22.3	19.7	(22)	
	5%	DB	22.1	23.0	27.0	32.0	37.1	39.1	39.9	39.8	37.9	34.2	29.8	24.2	(23)	
		MCWB	17.5	17.1	18.1	20.1	21.9	23.1	24.7	26.6	26.0	24.5	22.2	19.5	(24)	
	10%	DB	21.1	21.9	25.2	30.5	35.8	38.0	38.8	38.8	36.9	33.2	28.7	23.2	(25)	
		MCWB	17.0	17.1	18.2	20.4	22.2	23.8	25.5	27.2	26.3	24.7	22.1	18.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.5	20.5	22.1	24.3	28.0	29.7	31.7	32.1	31.1	28.8	25.8	22.7	(27)	
		MCDB	22.2	22.8	26.0	29.8	32.3	34.2	35.4	36.3	35.5	32.8	29.1	24.8	(28)	
	2%	WB	19.5	19.5	21.1	23.5	26.8	28.7	31.0	31.4	30.2	27.9	24.7	21.5	(29)	
		MCDB	21.6	22.0	24.4	29.0	31.9	33.6	35.1	35.8	34.9	32.1	28.5	23.8	(30)	
	5%	WB	18.5	18.9	20.4	22.8	26.0	27.8	30.3	30.9	29.6	27.2	23.9	20.6	(31)	
		MCDB	21.0	21.4	24.0	28.4	31.7	33.2	34.7	35.4	34.5	31.5	27.9	23.2	(32)	
	10%	WB	17.6	18.2	19.6	22.2	25.1	27.1	29.7	30.5	29.0	26.6	23.1	19.7	(33)	
		MCDB	20.3	21.0	23.6	27.8	31.6	33.6	34.5	35.2	34.0	31.1	27.1	22.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.4	6.3	7.0	7.7	7.2	6.9	6.7	6.7	6.3	5.2	5.0	(35)	
		MCDBR	6.2	7.1	9.0	9.4	9.8	9.1	8.7	8.1	7.6	7.3	6.1	5.7	(36)	
	5% WB	MCWBR	3.5	3.6	4.2	4.6	5.3	5.8	6.2	5.6	5.2	4.6	4.2	3.5	(37)	
		MCWBR	5.1	5.7	6.9	8.0	8.4	8.1	7.2	6.8	6.6	6.6	5.7	5.0	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.509	0.557	0.619	0.678	0.669	0.630	0.676	0.633	0.559	0.520	0.522	0.492	(40)	
		taud	1.937	1.827	1.694	1.625	1.630	1.698	1.657	1.752	1.879	1.963	1.959	1.995	(41)	
	Ebn at Noon	Ebn at Noon	720	719	702	676	682	704	672	699	740	743	703	716	(42)	
		Edh at Noon	166	199	239	262	261	242	252	228	196	171	161	151	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	3.61	4.34	5.22	5.83	6.72	7.24	6.82	6.47	6.00	4.99	3.75	3.41	(44)	
	RadStd	RadStd	0.25	0.29	0.38	0.39	0.20	0.19	0.16	0.14	0.12	0.15	0.26	0.26	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.59	N/A	-1.44	+0.73	N/A	N/A	N/A	N/A	+236	+224	(46)
(47)	Regional (5 neighbors)	+0.51	N/A	-1.52	+0.70	N/A	N/A	N/A	N/A	+196	+191	(47)

Nomenclature: See separate page